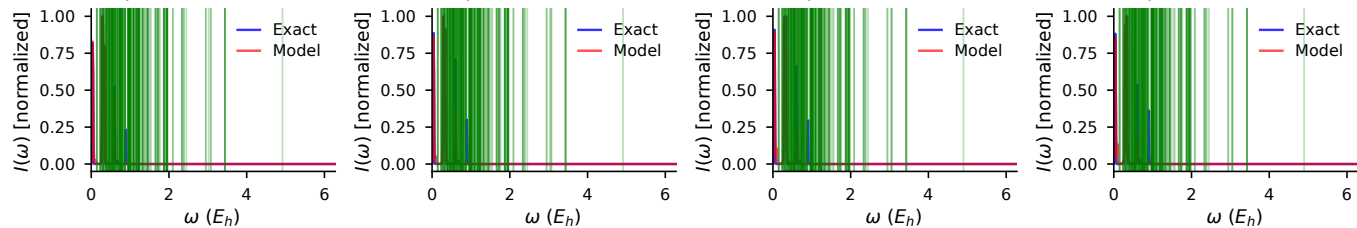
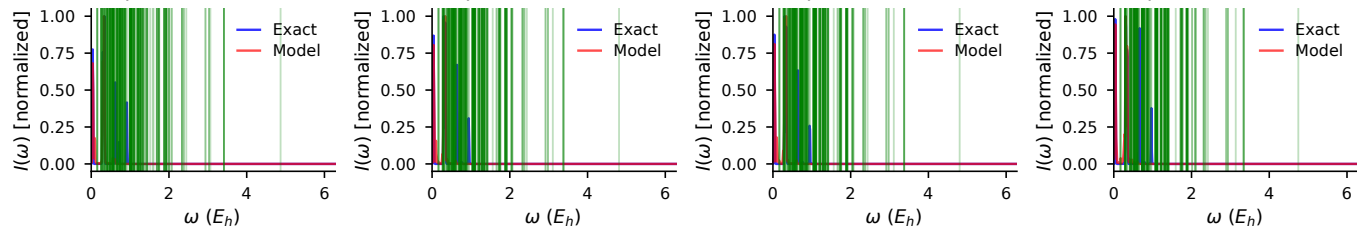


Spectral comparison: Model (red) vs Exact (blue), gaps (green)

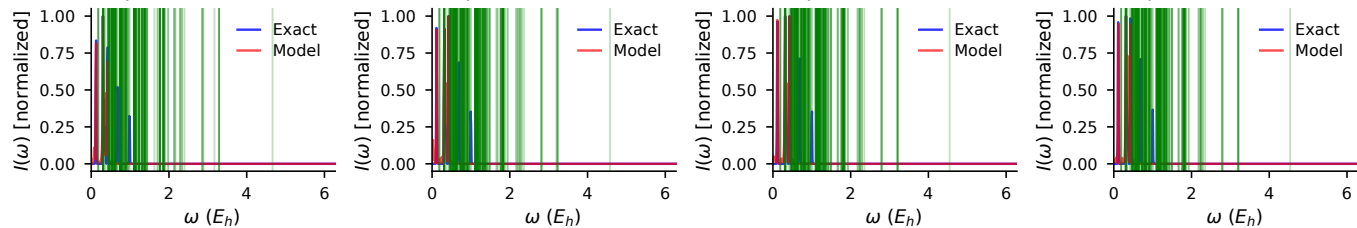
$R = 1.12$, $\text{AA} \setminus \text{quad}$ ($r = 1.00$, $\text{MSE} = 1.44 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 1.65 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 1.79 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 2.0 \times 10^{-6}$)



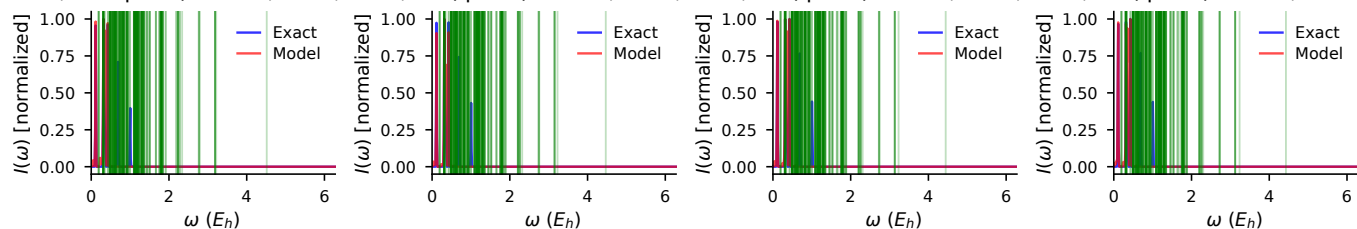
$R = 1.22$, $\text{AA} \setminus \text{quad}$ ($r = 1.00$, $\text{MSE} = 1.23 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 1.49 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 1.43 \times 10^{-6}$) quad ($r = 0.99$, $\text{MSE} = 2.3 \times 10^{-6}$)



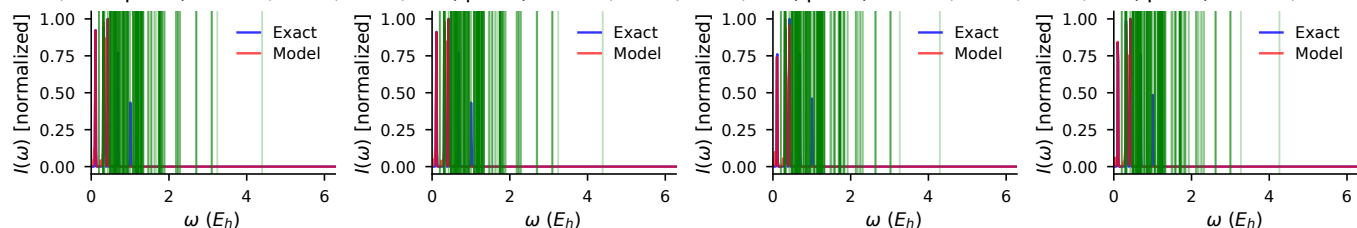
$R = 1.55$, $\text{AA} \setminus \text{quad}$ ($r = 0.98$, $\text{MSE} = 1.67 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 1.71 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 1.24 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 1.4 \times 10^{-6}$)



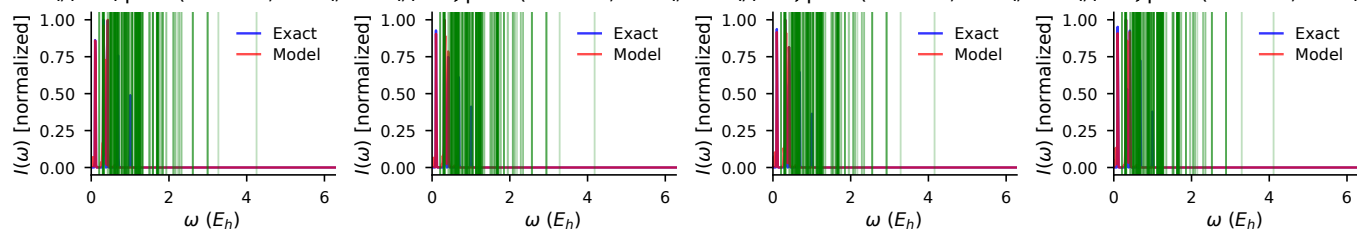
$R = 1.75$, $\text{AA} \setminus \text{quad}$ ($r = 1.00$, $\text{MSE} = 1.11 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 1.84 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 1.86 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 9.3 \times 10^{-7}$)



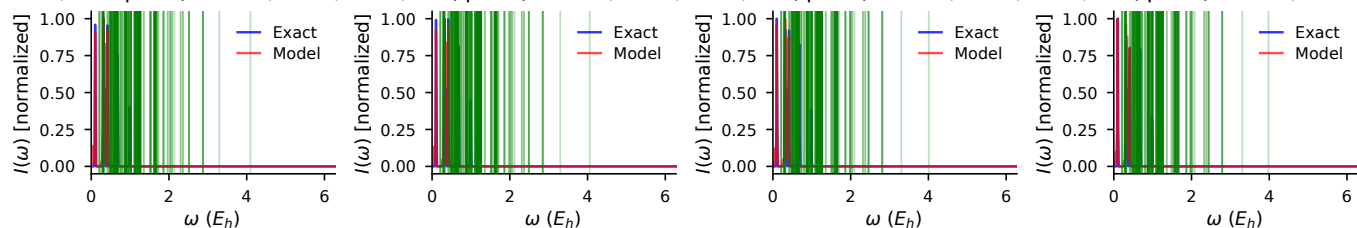
$R = 1.90$, $\text{AA} \setminus \text{quad}$ ($r = 1.00$, $\text{MSE} = 1.10 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 2.13 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 2.17 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 1.2 \times 10^{-6}$)



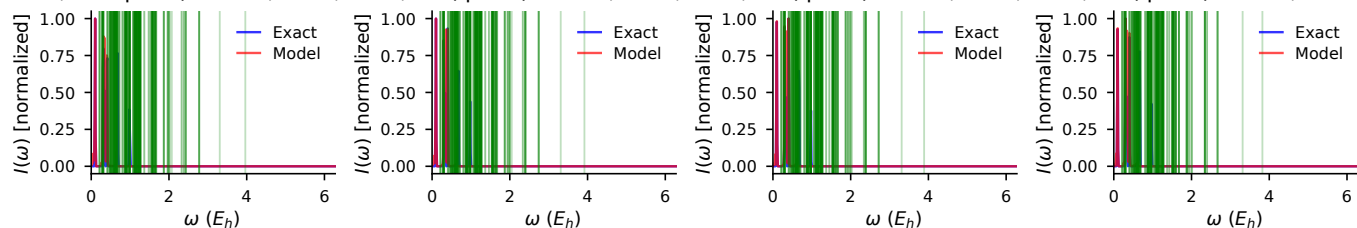
$R = 2.08$, $\text{AA} \setminus \text{quad}$ ($r = 1.00$, $\text{MSE} = 1.72 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 2.20 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 2.26 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 7.7 \times 10^{-7}$)



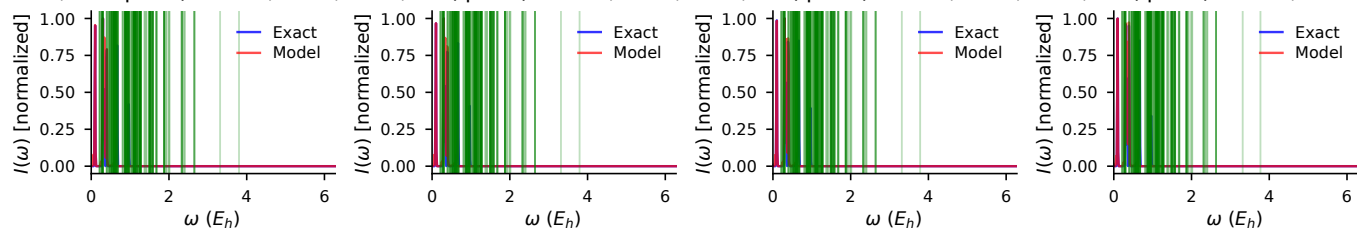
$R = 2.28$, $\text{AA} \setminus \text{quad}$ ($r = 1.00$, $\text{MSE} = 2.33 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 2.39 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 2.43 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 7.9 \times 10^{-7}$)



$R = 2.45$, $\text{AA} \setminus \text{quad}$ ($r = 1.00$, $\text{MSE} = 2.14 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 2.54 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 2.64 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 1.5 \times 10^{-6}$)



$R = 2.67$, $\text{AA} \setminus \text{quad}$ ($r = 1.00$, $\text{MSE} = 2.63 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 2.69 \times 10^{-6}$) quad ($r = 1.00$, $\text{MSE} = 2.71 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 8.0 \times 10^{-7}$)



$R = 2.74$, $\text{AA} \setminus \text{quad}$ ($r = 1.00$, $\text{MSE} = 2.78 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 2.81 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 3.80 \times 10^{-7}$) quad ($r = 1.00$, $\text{MSE} = 1.7 \times 10^{-6}$)

